BUSINESS INTELLIGENCE LAB

EXPERIMENT 5

Report Generation using Relational and Multidimensional Data using Microsoft Power BI

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1. Aim

To generate relational and multidimensional reports using Microsoft Power BI by utilizing transactional and dimensional data models, enabling effective business analysis, reporting, and decision-making.

2. Theory

Business Intelligence (BI): Business Intelligence transforms raw business data into meaningful information through data preparation, analysis, reporting and visualization.

Relational Reporting: Relational reports present detailed transactional records in tabular form. They are useful for operational analysis, record-level inspection, sorting and filtering.

Multidimensional Reporting: Multidimensional reports summarize data across dimensions such as geography, product and time. OLAP operations such as drill-down and drill-up help users move between summary and detailed levels.

Microsoft Power BI: Power BI provides Power Query for data preparation, a data model for organizing data, DAX for calculations, and visuals such as tables, matrices, charts, cards and slicers.

3. Tools / Software Required

- Microsoft Power BI Desktop
- Global Superstore Dataset.xlsx
- Power Query Editor
- DAX (Data Analysis Expressions)

4. Dataset Details

Dataset: Global Superstore Dataset (Kaggle)

Source: https://www.kaggle.com/datasets/apoorvaappz/global-super-store-dataset

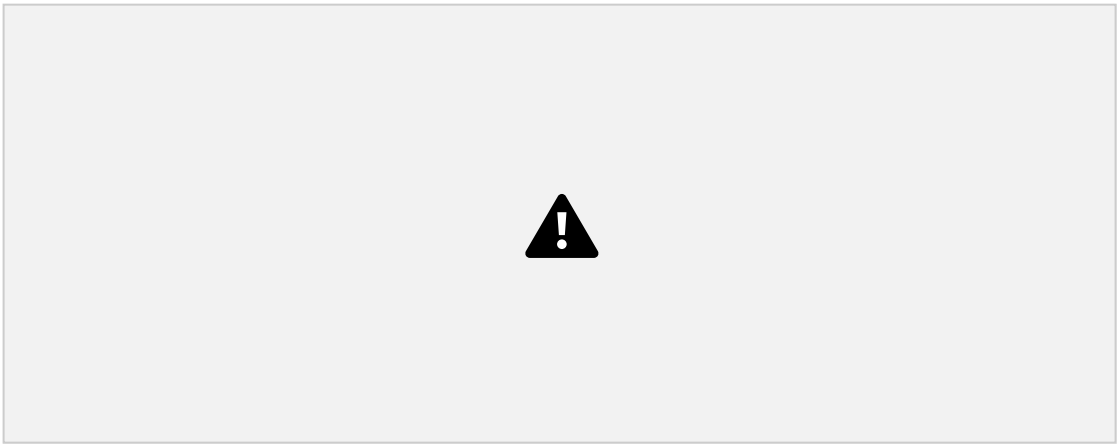
Important fields used: Order ID, Order Date, Customer Name, Product Name, Region, State, City, Category, Sales, Profit and Quantity. Field names and available records may vary slightly depending on the workbook version.

5. Implementation and Results

Part A — Generate a Relational Report

A Table visual was created in Power BI using Order ID, Customer Name, Product Name, Region, Sales and Profit. The table was resized for readability. Sorting was applied using the Sales column, and a visual-level Region filter was used to view records for a selected region. The table data can be exported through the visual's More options (…) menu by selecting Export data.

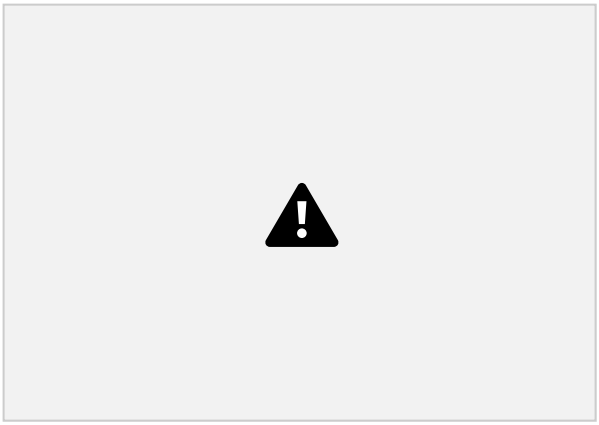
Observation: The relational report displays transaction-level details and helps inspect individual orders, customers, products, sales and profit. Sorting and filtering make it easier to locate relevant records.



Part B — Generate a Multidimensional Report

A hierarchy named Region Hierarchy was created in the order Region → State → City. A Matrix visual was inserted. Region Hierarchy was placed in Rows, while Sales, Profit and Quantity were placed in Values using Sum aggregation. The hierarchy was expanded and drill-down/drill-up operations were used to navigate between geographical levels.

Observation: The Matrix provides summarized business measures at different geographical levels. Drill-down reveals detailed state and city performance, while drill-up returns to higher-level regional summaries.

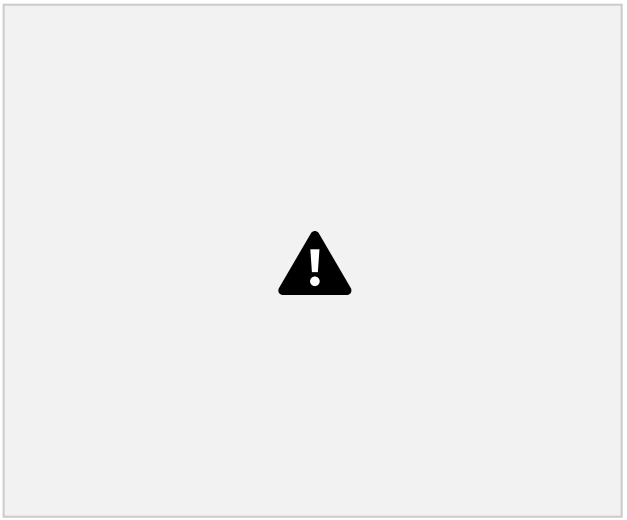


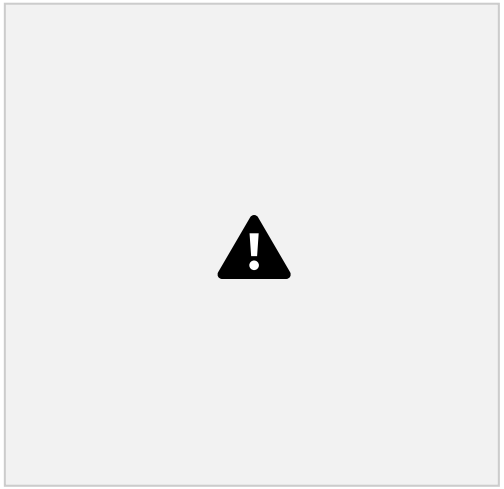
Part C — Comparative Analysis Report

Four visualizations were created to compare business performance:

Visualization	Fields / Configuration	Purpose
Sales by Category	Clustered column chart: Category vs. Sum of Sales	Compare sales across product categories.
Profit by Region	Clustered bar chart: Region vs. Sum of Profit	Compare profitability across regions.
Top 10 Customers by Sales	Bar chart: Customer Name vs. Sum of Sales; Top N = 10	Identify customers contributing the highest sales.
Year-wise Sales Trend	Line chart: Year from Order Date vs. Sum of Sales	Observe sales changes over time.

Observation: The comparative visuals provide different perspectives of business performance. Sales by Category compares product groups, Profit by Region compares geographical profitability, Top 10 Customers highlights major customers by sales, and the Year-wise Sales Trend shows changes in sales over time.





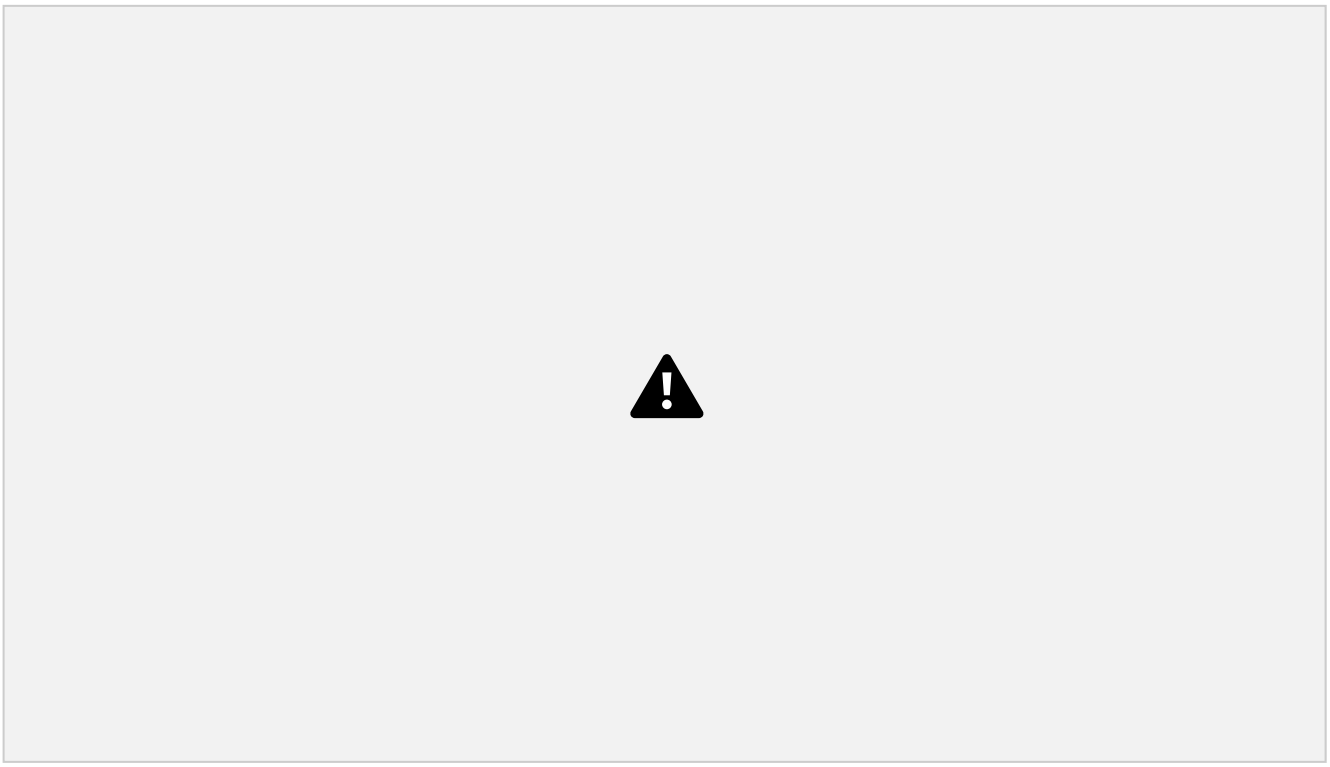
Part D — Interactive Dashboard

Three Card visuals were created: Total Sales using Sum of Sales, Total Profit using Sum of Profit, and Total Orders using Distinct count of Order ID. Three slicers were added for Region, Category and Year (using the Year level of Order Date). The visuals were arranged into a dashboard layout.

Each slicer was tested by selecting different values and checking whether the KPI cards and charts responded. If a visual did not respond, its interaction could be checked using Format → Edit interactions.

Important: The Total Orders card must use Distinct count of Order ID. An aggregation such as First Order ID displays an identifier rather than the total number of unique orders.

Observation: KPI cards summarize key business indicators, while slicers allow users to interactively filter the report and explore selected regions, categories and years.



6. Comparative Analysis: Relational vs. Multidimensional Reporting

Relational Report	Multidimensional Report
Displays individual or filtered transaction records.	Displays aggregated measures across dimensions.
Useful for operational inspection and record-level details.	Useful for comparisons, trends and management decisions.
Mainly uses tables, sorting and filters.	Uses matrices, hierarchies, aggregation and drill operations.

The relational approach answers questions about specific records, while the multidimensional approach helps identify broader patterns and compare performance across business dimensions.

7. Overall Observations

- The relational table provides detailed transactional information and supports sorting and filtering.
- The Matrix aggregates Sales, Profit and Quantity across the Region → State → City hierarchy.
- Drill-down and drill-up make it possible to move between summary and detailed geographical information.
- The four comparative charts provide category, region, customer and time-based views of business performance.
- KPI cards and slicers make the report easier to monitor and interact with.

8. Result

Relational and multidimensional reports were created in Microsoft Power BI using the Global Superstore dataset. Transaction-level reporting, geographical hierarchy analysis, comparative visualizations and an Interactive dashboard were prepared to support business analysis and decision-making

9. Conclusion

This experiment demonstrated how Microsoft Power BI can be used for detailed relational reporting as well as summarized multidimensional analysis. Tables, matrices, charts, KPI cards and slicers provide complementary ways to explore business data. Sorting, filtering, aggregation and drill operations help users interpret business performance and support informed decisions.

10. Learning Outcomes

- Generate relational reports using transactional business data.
- Create multidimensional reports using hierarchies and OLAP concepts.
- Apply filtering, grouping, aggregation and drill operations in reports.
- Design interactive dashboards using Power BI visualizations.
- Interpret business insights from different reporting approaches.